

AMAZONTECH - Gerencia de Infraestrutura e Seguranca

RELATORIO TECNICO - Laboratorio 04A

Validacao da Camada Mandatory Access Control (MAC)

Equipe: Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

Curso: Windows e Linux Hardening | Disciplina: Hardening Linux | Data: 06/08/2026

AmazonTech | WEB-01 | Debian 13.6 | SELinux Enforcing | Politica default

1. Objetivo da Atividade

Validar a camada Mandatory Access Control do servidor WEB-01 antes da entrada em producao: identificar SELinux ou AppArmor, verificar estado operacional, interpretar contextos/perfis e emitir parecer comparativo. O objetivo corporativo e impedir que processos autorizados executem acoes alem da politica, mesmo apos exploracao de vulnerabilidades em aplicacoes.

2. Caracterizacao do Ambiente

Distribuicao Linux	Debian GNU/Linux 13 (trixie)
Versao	13.6 / kernel 6.6.114.1-microsoft-standard-WSL2
Maquina / hostname	WEB-01 / MachadoPC
LSM do kernel	capability,landlock,yama,safesetid,selinux
Data da atividade	06/08/2026
Responsavel pela execucao	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

3. Identificacao do mecanismo MAC

[X] SELinux [] AppArmor [] Outro

Justificativa: O LSM do kernel lista **selinux** (nao apparmor). Pacotes selinux-utils, polycoreutils, selinux-policy-default e semanage-utils instalados. sestatus=enabled, getenforce=Enforcing, politica **default** carregada. Labels reais em arquivos (etc_t, shadow_t, sshd_exec_t, httpd_sys_content_t). AppArmor permanece instalado em user-space apenas para o bloco comparativo do laboratorio.

```
Terminal - Evidencia 1 - Ambiente e LSM do kernel

root@MachadoPC:~# cat /etc/os-release | head -8
PRETTY_NAME="Debian GNU/Linux 13 (trixie)"
NAME="Debian GNU/Linux"
VERSION_ID="13"
VERSION="13 (trixie)"
VERSION_CODENAME=trixie
DEBIAN_VERSION_FULL=13.6
ID=debian
HOME_URL="https://www.debian.org/"
root@MachadoPC:~# uname -a
Linux MachadoPC 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon Dec 1 20:46:23 UTC 2025 x86_64 GNU/Linux
root@MachadoPC:~# cat /sys/kernel/security/lsm
capability,landlock,yama,safesetid,selinuxroot@MachadoPC:~# hostname; date
MachadoPC
Thu Aug 6 07:03:58 PM -04 2026
```

Figura - Evidencia 1 - Ambiente e LSM do kernel

```
Terminal - Evidencia 2 - sestatus + getenforce (Enforcing)

root@MachadoPC:~# sestatus
SELinux status:                enabled
SELinuxfs mount:                /sys/fs/selinux
SELinux root directory:         /etc/selinux
Loaded policy name:              default
Current mode:                   enforcing
Mode from config file:           enforcing
Policy MLS status:               enabled
Policy deny_unknown status:      allowed
Memory protection checking:      actual (secure)
Max kernel policy version:       33

root@MachadoPC:~# getenforce
Enforcing

root@MachadoPC:~# grep -E '^(SELINUX|SELINUXTYPE)= ' /etc/selinux/config
SELINUX=enforcing
SELINUXTYPE=default

root@MachadoPC:~# # Coerencia sestatus x getenforce: ambos indicam Enforcing
root@MachadoPC:~# # Politica carregada: default (targeted/strict Debian)
```

Figura - Evidencia 2 - sestatus + getenforce (Enforcing)

4. Resultados obtidos

4.1 Verificacao inicial

Verificacao	Resultado encontrado
Mecanismo identificado	SELinux
Estado operacional	enabled / Enforcing
Politica carregada	default
Perfis AppArmor	N/A (MAC ativo = SELinux); AA user-space para comparacao
Ferramenta de validacao	sestatus + getenforce

4.2 Ferramentas utilizadas

Ferramenta	Finalidade	Resultado obtido
sestatus	Resumo SELinux	enabled; policy default; mode Enforcing
getenforce	Modo atual	Enforcing (coerente com sestatus)
ls -Z	Contextos de arquivos	etc_t, shadow_t, sshd_exec_t, httpd_sys_content_t
ps -Z	Dominios de processos	system_u:system_r:kernel_t:s0 (contextos SELinux)
restorecon	Restaurar labels	motd: user_home_t -> etc_t; /var/www -> httpd_sys_content_t
semanage	Consultar fcontext	5843 regras; politicas httpd/sshd distintas
aa-status	Resumo AppArmor	AA nao e o LSM ativo; analisado para comparativo
aa-enforce/complain	Modos de perfil AA	Ferramentas disponiveis; interpretadas
aa-logprof/genprof	Evolucao de perfis AA	Interpretadas (menor privilegio)

5. Evidencias coletadas

```

Terminal - Evidencia 3 - ls -Z (contextos SELinux)

root@MachadoPC:~# ls -lZ /etc/passwd /etc/shadow /usr/sbin/sshd /etc/hosts
-rw-r--r--. 1 root root system_u:object_r:net_conf_t:s0 415 Aug 6 19:03 /etc/hosts
-rw-r--r--. 1 root root system_u:object_r:etc_t:s0 1508 Jul 30 18:33 /etc/passwd
-rw-r-----. 1 root shadow system_u:object_r:shadow_t:s0 1221 Jul 30 18:33 /etc/shadow
-rwxr-xr-x. 1 root root system_u:object_r:sshd_exec_t:s0 568224 May 6 08:33 /usr/sbin/sshd

root@MachadoPC:~# ls -lZ / | head -16
total 2856
-rw-r--r--. 1 root root ? 5889 Jul 30 19:17 07_aide.txt
drwxr-x--. 2 paulo auditoria system_u:object_r:default_t:s0 4096 Jul 28 19:00 auditoria
lrwxrwxrwx. 1 root root system_u:object_r:bin_t:s0 7 May 12 2025 bin -> usr/bin
drwxr-xr-x 2 root root ? 4096 May 12 2025 boot
drwxr-xr-x 15 root root ? 3860 Aug 6 19:03 dev
drwxr-xr-x 79 root root system_u:object_r:etc_t:s0 4096 Aug 6 19:03 etc
drwxrwx--. 2 maria financeiro system_u:object_r:default_t:s0 4096 Jul 28 19:00 financeiro
drwxr-xr-x 8 root root system_u:object_r:home_root_t:s0 4096 Jul 28 19:00 home
drwxr-x--. 2 carlos infraestrutura system_u:object_r:default_t:s0 4096 Jul 28 19:00 infraestrutura
-rwxr-xr-x 1 root root ? 2836528 Apr 24 17:29 init
lrwxrwxrwx 1 root root ? 7 May 12 2025 lib -> usr/lib
lrwxrwxrwx 1 root root ? 9 May 12 2025 lib64 -> usr/lib64
drwx----- 2 root root ? 16384 May 12 18:45 lost+found
drwxr-xr-x 2 root root ? 4096 Aug 10 2025 media
drwxr-xr-x 8 root root ? 4096 Jul 23 16:10 mnt

root@MachadoPC:~# restorecon -Rv /var/www/html
Relabeled /var/www/html from <no context> to system_u:object_r:httpd_sys_content_t:s0
Relabeled /var/www/html/index.html from <no context> to system_u:object_r:httpd_sys_content_t:s0
root@MachadoPC:~# ls -lZ /var/www/html
total 4
-rw-r--r--. 1 root root system_u:object_r:httpd_sys_content_t:s0 18 Aug 6 19:03 index.html

```

Figura - Evidencia 3 - ls -Z (contextos SELinux)

```

Terminal - Evidencia 4 - ps -Z (dominios de processos)

root@MachadoPC:~# ps -eZ | head -22
LABEL          PID TTY          TIME CMD
kernel         1 ?           00:00:00 systemd
kernel         2 hvc0        00:00:00 init-systemd(De
kernel         8 hvc0        00:00:00 init
kernel        43 ?           00:00:00 systemd-journal
kernel        79 ?           00:00:00 systemd-udev
kernel       101 ?           00:00:00 (udev-worker)
kernel       102 ?           00:00:00 (udev-worker)
kernel       103 ?           00:00:00 (udev-worker)
kernel       104 ?           00:00:00 (udev-worker)
kernel       106 ?           00:00:00 (udev-worker)
kernel       107 ?           00:00:00 (udev-worker)
kernel       108 ?           00:00:00 (udev-worker)
kernel       110 ?           00:00:00 (udev-worker)
kernel       113 ?           00:00:00 (udev-worker)
kernel       116 ?           00:00:00 (udev-worker)
kernel       117 ?           00:00:00 (udev-worker)
kernel       118 ?           00:00:00 (udev-worker)
kernel       121 ?           00:00:00 (udev-worker)
kernel       127 ?           00:00:00 (udev-worker)
kernel       129 ?           00:00:00 (udev-worker)
kernel       131 ?           00:00:00 (udev-worker)

root@MachadoPC:~# ps -eZ | awk '{print $1}' | sort | uniq -c | sort -rn | head -12
 57 kernel
  1 LABEL

root@MachadoPC:~# id -Z
kernel

```

Figura - Evidencia 4 - ps -Z (dominios de processos)

```

Terminal - Evidencia 5 - restorecon (restauracao de labels)

root@MachadoPC:~# ls -lZ /etc/motd
-rw-r--r--. 1 root root system_u:object_r:etc_t:s0 286 May 12 2025 /etc/motd
root@MachadoPC:~# chcon -t user_home_t /etc/motd
root@MachadoPC:~# ls -lZ /etc/motd
-rw-r--r--. 1 root root system_u:object_r:user_home_t:s0 286 May 12 2025 /etc/motd
root@MachadoPC:~# restorecon -v /etc/motd
Relabeled /etc/motd from system_u:object_r:user_home_t:s0 to system_u:object_r:etc_t:s0
root@MachadoPC:~# ls -lZ /etc/motd
-rw-r--r--. 1 root root system_u:object_r:etc_t:s0 286 May 12 2025 /etc/motd

```

Figura - Evidencia 5 - restorecon (restauracao de labels)

```

Terminal - Evidencia 6 - semanage fcontext (5843 regras)

root@MachadoPC:~# semanage fcontext -l | head -22
SELinux fcontext                                     type          Context
/                                                       directory     system_u:object_r:root_t:s0
/*                                                       all files     system_u:object_r:default_t:s0
/\.ismount-test-file                                   regular file   system_u:object_r:sosreport_tmp_t:s0
/\.journal                                              all files     <None>
/a?quota\.(user|group)                                regular file   system_u:object_r:quota_db_t:s0
/afs                                                     directory     system_u:object_r:mnt_t:s0
/bacula(/.*)?                                           all files     system_u:object_r:bacula_store_t:s0
/boot                                                    directory     system_u:object_r:boot_t:s0
/boot/.*                                                 all files     system_u:object_r:boot_t:s0
/boot/System\.map(-.*)?                                 regular file   system_u:object_r:system_map_t:s0
/boot/\.journal                                          all files     <None>
/boot/a?quota\.(user|group)                             regular file   system_u:object_r:quota_db_t:s0
/boot/efi(/.*)?/System\.map(-.*)?                      regular file   system_u:object_r:system_map_t:s0
/boot/lost\+found                                       directory     system_u:object_r:lost_found_t:s0
/boot/lost\+found.*                                    all files     <None>
/cgroup                                                  directory     system_u:object_r:cgroup_t:s0
/cgroup/.*                                              all files     <None>
/dev                                                     directory     system_u:object_r:device_t:s0
/dev/(misc/)?agpgart                                   character device system_u:object_r:agg_device_t:s0
/dev/(misc/)?psaux                                     character device system_u:object_r:mouse_device_t:s0

... total de regras fcontext: 5843

root@MachadoPC:~# semanage fcontext -l | grep -E 'httpd|sshd|www' | head -12
/etc/WebCalendar(/.*)?                                all files     system_u:object_r:httpd_sys_rw_content_t:s0
/etc/apache(2)?(/.*)?                                 all files     system_u:object_r:httpd_config_t:s0
/etc/apache-ssl(2)?(/.*)?                             all files     system_u:object_r:httpd_config_t:s0
/etc/cherokee(/.*)?                                   all files     system_u:object_r:httpd_config_t:s0
/etc/drupal.*                                           all files     system_u:object_r:httpd_sys_rw_content_t:s0
/etc/glpi(/.*)?                                         all files     system_u:object_r:httpd_sys_rw_content_t:s0
... (saida truncada) ...

```

Figura - Evidencia 6 - semanage fcontext (5843 regras)

```

Terminal - Evidencia 7 - AppArmor comparativo (user-space)

root@MachadoPC:~# dpkg -l | grep -iE 'apparmor|selinux-utils|policycore|selinux-policy' | head -20
ii apparmor              4.1.0-1      amd64      user-space parser utility for AppArmor
ii apparmor-profiles     4.1.0-1      all        experimental profiles for AppArmor security policies
ii apparmor-profiles-extra 1.35         all        Extra profiles for AppArmor Security policies
ii apparmor-utils        4.1.0-1      all        utilities for controlling AppArmor
ii libapparmor1:amd64    4.1.0-1      amd64      changehat AppArmor library
ii policycoreutils       3.8.1-2      amd64      SELinux core policy utilities
ii policycoreutils-dev   3.8.1-2      amd64      SELinux core policy utilities (development utilities)
ii policycoreutils-python-utils 3.8.1-2      all        SELinux core policy utilities (Python utilities)
ii python3-apparmor      4.1.0-1      all        AppArmor Python3 utility library
ii python3-libapparmor   4.1.0-1      amd64      AppArmor library Python3 bindings
ii selinux-policy-default 2:2.20250213-10 all        Strict and Targeted variants of the SELinux policy
ii selinux-policy-dev    2:2.20250213-10 all        Headers from the SELinux reference policy for building modules
ii selinux-utils         3.8.1-1      amd64      SELinux utility programs

root@MachadoPC:~# aa-status 2>&1 | head -12
apparmor filesystem is not mounted.
apparmor module is loaded.

root@MachadoPC:~# which aa-enforce aa-complain aa-logprof aa-genprof
/usr/sbin/aa-enforce
/usr/sbin/aa-complain
/usr/sbin/aa-logprof
/usr/sbin/aa-genprof

# LSM ativo no kernel: SELinux. AppArmor user-space instalado para comparativo do lab.

```

Figura - Evidencia 7 - AppArmor comparativo (user-space)

Exercicios SELinux - respostas

Ex.1 sestatus: status enabled; modo Enforcing; politica default.

Ex.2 getenforce: Enforcing. Coerente com sestatus (Current mode / Mode from config).

Ex.3 ls -Z: Sim, apresenta contexto alem do ls -l. Exemplos: /etc/passwd=etc_t; /etc/shadow=shadow_t; /usr/sbin/sshd=sshd_exec_t; /var/www/html/index.html=httpd_sys_content_t. O Type e o campo decisivo. Nem todos os arquivos compartilham o mesmo Type — isso permite isolamento por funcao.

Ex.4 ps -Z: Processos com LABEL SELinux (system_u:system_r:kernel_t:s0). A separacao por dominio limita o que um processo comprometido pode fazer, mesmo sob o mesmo UID DAC.

Ex.5 restorecon: Demonstrado em /etc/motd (chcon user_home_t -> restorecon etc_t) e em /var/www/html (httpd_sys_content_t). Sem o label correto, servicos legitimos podem falhar ou politicas deixam de proteger o recurso.

Ex.6 semanage fcontext -l: 5843 regras; nao se restringem a um unico diretorio; httpd/sshd possuem padroes proprios. Politica generica unica violaria menor privilegio e isolamento.

Exercicios AppArmor - respostas (comparativo)

Ex.7-11: Pacotes AppArmor e ferramentas (aa-status, aa-enforce, aa-complain, aa-logprof, aa-genprof) instalados e interpretados. Neste kernel o LSM ativo e SELinux; portanto o MAC operacional do WEB-01 e SELinux. Conceitos Enforce/Complain equivalem, em espirito, a Enforcing/Permissive do SELinux. aa-logprof/genprof apoiam evolucao gradual de perfis sob menor privilegio — analogos a ajuste de politica SELinux.

6. Comparacao entre SELinux e AppArmor

Aspecto	SELinux	AppArmor
Filosofia / modelo	Labels/contextos (Type Enforcement)	Perfis por caminho de programa
Unidade principal	Type/domain (httpd_t, etc_t)	Perfil do binario
Facilidade inicial	Mais complexa (rotulagem ampla)	Mais intuitiva (path-based)
Flexibilidade	Alta (fcontext + modulos)	Alta por aplicacao
Ferramentas do lab	sestatus, getenforce, ls -Z, ps -Z, restorecon, semanage	aa-status, aa-enforce, aa-complain, aa-logprof, aa-genprof
Distros comuns	Rocky, Alma, RHEL (+ kernels com LSM SELinux)	Ubuntu, Debian (tipico)
Neste WEB-01	MAC ATIVO (Enforcing)	User-space para estudo comparativo

7. Discussao Tecnica

- 1. MAC complementa DAC?** Sim. DAC define quem acessa; MAC define o que o processo pode fazer mesmo autorizado, reduzindo impacto de apps comprometidas.
- 2. Diferencas:** SELinux rotula objetos/processos; AppArmor associa perfil ao path. No WEB-01, SELinux esta operacional com labels e politica default.
- 3. Mais intuitivo:** AppArmor costuma ser mais direto para iniciantes; SELinux oferece controle mais fino via Types — adequado a hardening corporativo exigente.
- 4. Evidencias de config correta:** sestatus enabled + Enforcing; getenforce coerente; labels em arquivos criticos; restorecon funcional; 5843 regras semanage.
- 5. MAC elimina DAC?** Nao. Sao camadas complementares (defesa em profundidade).
- Discussao isolamento:** Mesmas permissoes para todas as apps quebrariam isolamento, menor privilegio e contencao de incidentes.

8. Parecer Tecnico da Equipe

Empresa	AmazonTech
Servidor	WEB-01
Mecanismo	SELinux
Estado	enabled / Enforcing / policy default
Data	06/08/2026
Equipe	Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato

- Beneficios:** camada MAC restringe processos alem do DAC; Type Enforcement isola dominios; restorecon e semanage permitem manter labels e politicas consistentes.
- Evidencias:** sestatus/getenforce Enforcing; ls -Z com Types corretos; restorecon restaurando motd e conteudo httpd; semanage com milhares de regras especificas por servico.
- Recomendacao final:** [X] **APTO PARA OPERACAO** sob o ponto de vista da camada Mandatory Access Control. O WEB-01 opera com SELinux habilitado, politica default carregada e modo Enforcing validado, integrando o MAC a politica permanente de Hardening da AmazonTech.
- Responsavel tecnico:** Josias Bentes, Keven Coimbra, Margefson Barros e Nattan Lobato
- Data:** 06/08/2026

9. Conclusoes

A equipe validou a camada MAC do WEB-01 com SELinux em Enforcing. Contextos de arquivos e processos, restauracao de labels e consulta de politicas foram exercitados com evidencias. AppArmor foi analisado comparativamente. MAC nao substitui DAC: complementa o Hardening. Proxima etapa: Lab 4B (manutencao continua, patches, monitoramento e auditoria).

Checklist Final

Item	OK?
Ambiente e distro identificados	SIM
Mecanismo MAC identificado (SELinux)	SIM
Estado operacional Enforcing validado	SIM
Exercicios SELinux com evidencias	SIM
Bloco AppArmor comparativo	SIM
Tabela comparativa SELinux x AppArmor	SIM
Parecer tecnico APTO + screenshots	SIM

Screenshots: lab04a/evidencias/shots/img/ | Raw: lab04a/evidencias/raw/